

EBCPA: Efficient Blockchain-Based Conditional Privacy-Preserving Authentication for VANETs

Chao Lin^{ID}, Xinyi Huang^{ID}, and Debiao He^{ID}

Abstract—Vehicular Ad-hoc Networks (VANETs) are with great potentials to facilitate traffic management and improve driver safety. Blockchain-based conditional privacy-preserving authentication (BCPPA) is proposed to achieve an optimal tradeoff among anonymity, traceability and key/certificate management in VANETs. Existing BCPPA protocols mitigate these security and privacy challenges by adding a significant cost on verification and traceability. As a result, current solutions fail to meet high mobility, low latency, and real-time requirements of VANETs. In this article, we design three new system building blocks namely key derivation (KeyDer), signatures of knowledge (SoK) and smart contract, following by a more efficient BCPPA protocol (named as EBCPA). To show the advantage of EBCPA, we first demonstrate it can satisfy the necessary requirements (e.g., message authentication, conditional privacy protection, resilience to common attacks, and so forth). Moreover, we implement the EBCPA in the on-line Ethereum test network (Rinkeby), Hyperledger test network and VANETs simulation environment (via VanetMobiSim and NS-2). Finally, we evaluate its communication overhead and computational cost via comparing to existing BCPPA protocols that strive to achieve similar properties. From the implementation and comparison results, our proposal can improve efficiency by reducing the time cost of traceability at least 48.95% and verification at least 42.21%.

Index Terms—Vehicular ad-hoc networks (VANETs), conditional privacy-preserving authentication (CPPA), blockchain, signatures of knowledge, key derivation

1 INTRODUCTION

THE rapid increase of vehicles and the continuous development of information technology (e.g., automatic driving, 5G and network slicing) have drawn wide attention in vehicular ad-hoc networks (VANETs). In VANETs, vehicles

can collect various messages including velocity, location and weather condition via global position system (GPS), radio frequency identification devices (RFID), electronic sensors or automobile data recorders [1], [2]. These messages are shared among vehicles and roadside units (RSUs) for responding to real-time changes such as avoiding collision, intelligent routing and traffic lighting, and consequently this will greatly enhance driving safety and convenience [3].

Typical network framework of VANETs (Fig. 1) mainly comprises five entities, namely traffic control center (TCC), RSU, vehicle, on-board unit (OBU) and Internet, as well as three communication types, namely wired/wireless Internet, vehicle-to-vehicle (V2V), and vehicle-to-RSU (V2R). OBUs are embedded into vehicles for communicating with other vehicles (or RSUs), to share real-time traffic information via wireless V2V (resp. wireless V2R). TCC is responsible for taking timely actions (e.g., adjusting traffic lights) according to traffic messages obtained from RSUs [4].

While the above framework can improve smart processing and real-time response in modern intelligent transportation, public wireless communications of V2V and V2R make VANETs vulnerable to various attacks [5]. Specifically, attackers may create social chaos via intercepting, tampering, or deleting transmitting data. As an example, millions of General Motors (GM) cars and trucks were vulnerable to a remote exploit for nearly half a decade, which can track vehicles, or engage their brakes at high speed, or even disable brakes altogether¹.

Authentication can mitigate the above issues, but it poses a privacy risk to vehicles and corresponding drivers. The real identity contained in transmitting messages may be

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1. <https://www.synopsys.com/glossary/what-is-connected-car-cyber-security.html>

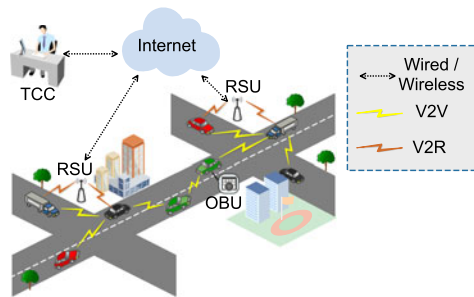


Fig. 1. Typical network framework of VANETs.

abused by attackers to trace more private information including driver name, home address and company information. For instance, according to the survey data that most vehicles are in parked status 95% of a day average, attackers can effectively deduce the personal activities of a targeted vehicle via analyzing its transmitting traffic data such as the destination, parking lots and interval time [6]. This may seriously threaten the property security or even drivers' personal safety, and hence we should highlight the importance of anonymity protection in VANETs.

On the other hand, the absolute anonymity may lead to malicious message issuing, namely, malicious vehicles may share false traffic status such as the number of vehicles, vehicle speed or traffic lights, to confuse other vehicles or cause a traffic jam/accident. Thus, a traceability mechanism is indispensable to track these malicious behaviors. To deal with these issues, conditional privacy-preserving authentication (CPPA) has been proposed [7], [8], which not only ensures the authenticity, validity and integrity of transmitting messages, but also provides the anonymity of honest vehicles and traceability of malicious behaviors.

Existing CPPA protocols could be broadly divided into four types, namely PKI-based [7], [8], ID-based [9], [10], certificateless [11] [12], and blockchain-based [13], [14]. The PKI-based solutions are generally effective and simple to realize, but most of them involve significant storage costs (i.e., a large number of key/certificates are pre-loaded into OBUs for ensuring anonymity) and intractable certificate managements (e.g., issuing, updating and revocation). In the ID-based solutions, vehicles' secret keys are generated by a trusted entity (e.g., private key generator, PKG) upon the vehicles' identities. This can avoid the complex key/certificate storage and management issue faced by the PKI-based solutions. Additionally, ID-based solutions like [15], [16], [17] can further provide efficient batch verification.

Nevertheless, existing ID-based solutions suffer from the issues of key escrow and key revocation. The key escrow refers to that a vehicle's secret key is generated by the PKG, who also knows this secret key. Once the PKG is compromised, it can impersonate any vehicle to share counterfeit traffic messages and hence causes serious disturbance. This issue is then solved by the certificateless CPPA, where the vehicle's secret key is determined by both the PKG (also named as Key Generation Center (KGC)) and itself.

However, in both ID-based and certificateless CPPA, the key revocation issue is still unsolved. The vehicle's expired secret key cannot be revoked conveniently because the vehicle's identity is regarded as the public information in ID-based or certificateless solutions. The certificate revocation list of

PKI-based solutions is not suitable for ID-based or certificateless solutions, since such a list (if for maintaining revoked identities) will revoke the entire authority of a vehicle. However, the revocation in ID-based solutions generally refers to that those disclosed secret keys should be revoked and updated, but not the whole identity.

Blockchain, an emerging technology with decentralization, immutability, and verifiability, can potentially solve the above issues. That is, those public information (such as certificates, anonymous identities/public keys) can be managed via the blockchain smart contract, such that the authentication (especially verification) and revocation (deleting public information logically) can be efficiently realized. Also, it only requires retrieving public information from blockchain during the authentication, which is considerably efficient for involving no new data storage into blockchain.

However, most blockchain-based conditional privacy-preserving (BCPPA) protocols such as [13], [18], [19] involve high communication overhead, frequent interactions or trustworthy hardware (i.e., a device whose stored data cannot be extracted). Recently, to further ease these issues, Lin *et al.* [14] and Feng *et al.* [20] contributed two proposals with privacy-preserving authentication and traceability, but neither of them can efficiently verify message integrity or trace malicious behaviors. This motivates us to propose a more efficient and versatile BCPA protocol for VANETs.

Contributions. Our main contribution is a more efficient BCPA protocol (EBCPA) in VANETs. The EBCPA not only achieves a tradeoff among anonymity, traceability and key/certificate management, but also eliminates frequent interactions, costly computation or trustworthy hardwares.

- 1) The core idea of our EBCPA is a traceable one-time public key (also named as anonymous public key) generation mechanism. This mechanism can ensure vehicles to authenticate each other anonymously and only a trusted entity (i.e., manager) can recover the long-time public key, namely the real identity of a vehicle. The concrete construction of EBCPA is based on a reconstructed key derivation (KeyDer) scheme and the authentication is realized by a new signatures of knowledge (SoK) scheme.
- 2) Blockchain is employed to record anonymous public keys publicly, such that vehicles can adopt these keys to share traffic data. Due to the use of KeyDer, the anonymous public keys could be submitted in advance and periodically (e.g., pre-submitting about 240 anonymous public keys once for an interval of two days). The authentication between two vehicles only involves on-chain retrieval operations, which can be efficiently realized. This indeed mitigates the negative effects caused by the low throughput and high latency of blockchain.
- 3) In addition to the efficient secret key/certificate revocation and trustworthy hardware-free, our EBCPA protocol is more efficient than existing BCPA protocols especially in traceability and verification. This will be demonstrated via security analysis and performance evaluation. Specifically, our proposal improves the efficiency of traceability (at least 48.95%) and verification (at least 42.21%) than recent BCPA protocols (i.e., CertCoin [18], BCPA [14] and BPAS [20]).

Organization. The rest sections are organized as follows. Sections 2 and 3 review the related work of CPPA and involving preliminaries (including notations, cryptographic primitives, system model, and security/privacy requirements) respectively. In Section 4, we introduce design details of our KeyDer scheme, SoK scheme and smart contract, following by the constructed EBCPA protocol. Then, we demonstrate the advantage of our proposal through security analysis and performance evaluation in Sections 5 and 6 respectively. Section 7 is the conclusion of this paper.

2 RELATED WORK

The concept of CPPA in VANETs has been studied intensively, which can be broadly classified into PKI-based, ID-based, certificateless, and blockchain-based.

PKI-Based. In 2007, Raya and Hubaux [7] first introduced a PKI-based CPPA which can deal with security and privacy issues in VANETs. However, their proposal requires pre-loading lots of public/secret key pairs together with anonymous certificates. This will cause a considerable storage overhead and also complex operations for achieving key/certificate revocation. In addition, Freudiger *et al.* [21] and Zhang *et al.* [22] adopted mix-zones and k -anonymity respectively to propose new CPPA protocols. While their proposals are more efficient than previous solutions, they still involve costly certificate storages on RSUs and vehicles.

To address the above issue, Lu *et al.* [8] introduced RSU-based anonymous certificates into CPPA. That is, vehicles can request a temporary anonymous certificate from their nearby RSUs, such that they can anonymously communicate with each other. While such a mechanism indeed achieves the conditional privacy protection, the communication procedure (including signing and verification) is subject to the online RSUs. Worse still, these certificate-based CPPA protocols involve costly certificate managements.

ID-Based. ID-based CPPA protocols are proposed to eliminate the certificate management issue, which mainly adopt ID-based signature [23], [24], [25], software-based method [9], pseudo-ID-based method [15], and so forth.

These ID-based CPPA protocols can improve the security or efficiency of certificate-based solutions, but most of them are either requiring a trustworthy hardware or difficult to be applied into multi-cloud environment. To address the first challenge, the one-time identity-based aggregate signature and Chinese Remainder Theorem (CRT) have been used in [16] and [17] respectively (both of which only involve realistic tamper-proof devices). With regard to the second challenge, Cui *et al.* [26] employed the elliptic curve cryptography to propose a robust and extensible CPPA protocol to satisfy the need of growing diversified services. However, existing ID-based solutions are faced with the troublesome escrow/revocation of vehicles' secret keys.

Certificateless. Certificateless CPPA protocols were proposed to solve the key escrow issue in ID-based CPPA protocols. That is, an attacker is unable to obtain the vehicle's full secret key even it colludes with the KGC. In 2015, Horng *et al.* [11] proposed a new certificateless signature scheme for conditional privacy protection in VANETs, although their proposal was then shown vulnerable to malicious-but-passive key generation center attacks [27]. In 2018, Ming *et al.* [28] proposed a practical

certificateless CPPA protocol for VANETs, which does not involve intractable bilinear pairing and map-to-hash operations. Recently, Xu *et al.* [12] employed the CRT to propose a novel certificateless CPPA for VANETs with batch verification and conditional privacy protection. Existing certificateless CPPA protocols may have mitigated the privacy, security and efficiency issues in VANETs, but they are still faced with the intractable key revocation.

Blockchain-Based. Blockchain, is a distributed ledger technology, which consists of chronologically chained hashing blocks. This special structure, together with other technologies such as peer-to-peer network, cryptography and consensus, ensures the decentralization, immutability, and verifiability of blockchain. While blockchain has been classified into public, permissioned, private, all of them are still faced with insufficiencies such as low throughput and high latency [29], [30]. The potential of blockchain in VANETs is that we can realize the authentication with only on-chain retrieval operations, which can mitigate the negative effects caused by the insufficiencies of blockchain.

Blockchain has been studied to solve the aforementioned issues (i.e., non-transparency of trusted authorities and complex certificate revocation) existing in PKI-based CPPA protocols and the intractability of key escrow/revocation in ID-based or certificateless protocols. In 2019, Lu *et al.* [31] combined blockchain with Merkle Patricia Tree to design a blockchain-based privacy-preserving authentication protocol. Their proposal can protect the privacy of vehicles and revoke the certificates efficiently, but it involves multi-interactions between vehicles and CA for generating certificates. In 2020, Gabay *et al.* [32] integrated a token-based mechanism and Pederson commitment into blockchain to propose two anonymous authentication protocols. Other solutions such as [33] were also proposed for the privacy-preserving authentication in VANETs. However, none of these solutions took the traceability into account, failing to trace the malicious behavior of vehicles (e.g., issuing counterfeit traffic information).

In 2018, to further introduce traceability into the privacy protection for blockchain-enabled VANETs, Li *et al.* [18] employed a threshold ring signature to construct a CertCoin protocol. However, the efficiency and privacy of CertCoin are seriously affected by the ring and threshold size. Recently, Lin *et al.* [14] integrated blockchain with a key derivation algorithm to propose a BCPPA protocol, and Feng *et al.* [20] combined attribute-based encryption and blockchain-enabled smart contract to design a BPAS protocol. However, neither of them can achieve efficient traceability, where the former is linearly affected by the average searching number of one-time public keys and the latter is limited to the number of leaf nodes in the access tree. In addition, the efficiency of message verification in both [14] and [20] can be further improved. Thus, we are motivated to propose a more efficient BCPPA protocol for achieving a tradeoff among security, privacy and utility for VANETs.

3 PROBLEM DEFINITION

This section mainly presents involved notations and cryptographic primitives, together with defining system model and security/privacy requirements of BCPPA in VANETs.

3.1 Notations

Table 1 lists main involved notations throughout this paper.

TABLE 1
Involved Notations in This Paper

Notation	Description
λ	a security parameter.
p, q	two large prime numbers.
E	an elliptic curve defined by $y^2 = x^3 + ax + b \pmod{p}$, where $a, b \in \mathbb{F}_p$.
$\mathbb{G}$	an additive group with order q , comprising all points on E and an infinite point O .
P	a generator of $\mathbb{G}$.
$pseed$	a private seed in the key derivation scheme.
i, j	two positive integers.
pd_i^{root}	the i th vehicle's root private derivation information.
ask_i^j	the i th vehicle's j th deriving anonymous secret key.
$APK_i^{j,1}, APK_i^{j,2}$	the i th vehicle's j th deriving anonymous public key.
pd_i^j	the i th vehicle's j th private derivation information.
lsk_i	the i th vehicle's long-term secret key.
LPK_i	the i th vehicle's long-term public key.
lsk_m	the manager's long-term secret key.
LPK_m	the manager's long-term public key.
$\mathcal{H}$	a secure hash function $\mathcal{H} : \{0, 1\}^* \rightarrow \mathbb{Z}_q^*$.
msg_i	the i th vehicle's sharing traffic message.
ts_i	the i th vehicle's timestamp to share message.
$R = (x_i, w_i)$	a relation with instance x_i and corresponding witness w_i .
$\parallel$	concatenation operation of strings.
PPT	abbreviation of probabilistic polynomial time.
$negl(\cdot)$	a negligible function.
tx_rtv	retrieving transactions from blockchain.
eth_calls	performing a message call in Ethereum.
$E\text{Sign}$	signing algorithm of a secure digital signature scheme (e.g., ECDSA).
$E\text{Verf}$	corresponding verification algorithm of the secure digital signature scheme.

3.2 Cryptographic Primitives

This subsection briefly reviews the involved cryptographic primitives in our proposal, namely, Σ -protocols and signatures of knowledge.

3.2.1 Σ -Protocols

Σ -protocols [34] are with a 3-move form between a prover P and a verifier V , where P would like to convince V that it owns a witness w such that $(x, w) \in R$ without showing w directly. After the interaction, a triple (a, c, z) will be output, in which a and z are computed by P based on a random challenge c chosen by V . A Σ -protocol is required to satisfy three different properties, namely, *completeness* (i.e., (x, a, c, z) will be always accepted if there exist a verification algorithm ϕ such that $\Pr[\phi(x, a, c, z) = 1] \geq 1 - \text{negl}(\lambda)$, and this property is perfect if the probability is equal to 1), *special soundness* (i.e., given two accepted (x, a, c, z) and (x, a, c', z') where $c \neq c'$, the witness w can be efficiently computed) and *special honest-verifier zero-knowledge* (i.e., there exists a simulator S can output an accepted (x, a, c, z) based on a known c , and the output is indistinguishable from that generated by a real interaction between P and V).

Notably, a Σ -protocol could be turned into non-interactive via Fiat-Shamir heuristic [35]. That is, P first computes a and adopts a secure hash function $\mathcal{H}$ to compute $c = \mathcal{H}(x, a)$, then it executes the remaining operations of Σ -protocol to compute z and sends (x, a, c, z) to V . Then, we can prove the soundness and zero-knowledge of the new protocol in the random oracle (RO) model [36] by replacing $\mathcal{H}(x, a)$ with a random oracle [35].

3.2.2 Signatures of Knowledge

Signatures of knowledge [37] is a proof system that allows one entity to convince others about knowing a secret value

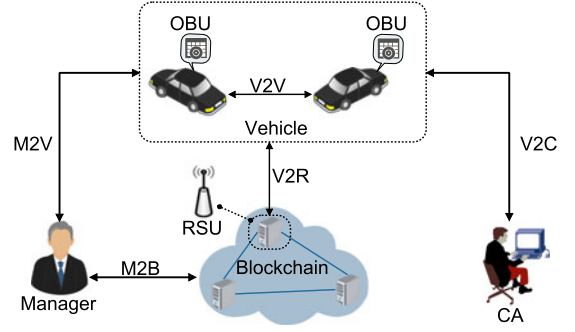


Fig. 2. Architecture of blockchain-enabled VANETs.

but without revealing it. Also, it has the functionality of digital signature, that is, ensuring message integrity and identity authenticity. Such a cryptographic primitive is generally transformed from a Σ -protocol on basic of Fiat-Shamir heuristic. Thus, the SoK can be proven secure in the random oracle model and its interactive version owns the zero-knowledge property. Specifically, a SoK scheme is comprised by the following three algorithms.

- **Setup(1^λ)**: This parameter generation algorithm takes a security parameter λ as input, and outputs a public parameter pp .
- **GenProof(pp, m, x, w)**: This proof algorithm takes public parameter pp , message m and a relation $(x, w) \in \mathcal{R}$ as inputs (where x is an instance and w is the secret witness), and outputs a SoK proof π .
- **VerfProof(pp, m, x, π)**: This verification algorithm takes public parameter pp , message m , instance x and SoK proof π as inputs, it outputs 1 if the π is valid; Otherwise it outputs 0.

3.3 System Model

In the system model of our EBCPA protocol, there are five participants, namely, *Certificate Authority (CA)*, *Vehicle*, *Road Side Units (RSU)*, *Blockchain* and *Manager* (see Fig. 2). Communication methods mainly involve M2B, M2V, V2V, V2R, and V2C, where M2B is for a manager to communicate with blockchain nodes (e.g., RSUs) for submitting/retrieving anonymous public keys, M2V is for vehicles to register their anonymous public keys with manager, V2V is for vehicles to communicate with each other via dedicated short range communications (DSRC) protocol, V2R is for vehicles to query transactions from the blockchain or to share traffic status with RSUs, and V2C is for vehicles/RSUs to register certificates from CA offline/online.

- **CA**: This trusted entity, with abundant computation and storage resources, mainly manages certificates of vehicles' or RSUs' public keys. In our proposal, the vehicles and RSUs need to register certificates from the CA before the authentication. Thus, the CA will generate and return certificates to the vehicles or RSUs offline/online. The certificate management can be realized based on blockchain or existing public key certificate system (e.g., X.509) [38].
- **Manager**: This trusted entity is in charge of registering vehicles' anonymous public keys through the smart contract. Then, the vehicles can adopt these registered

keys to communicate with other vehicles or RSUs anonymously. In our model, the manager is the only entity owning the ability to track real identities of targeted malicious vehicles.

- *Vehicle*: This entity is equipped with an internal processing unit (i.e., OBU) owning the tamper-proof property and supporting DSRC protocol. On basis of the OBU, vehicles are able to communicate with other vehicles/RSUs. It is worth noting that the OBU in our proposal is realistic, for its stored secret keys are updated periodically. Specifically, each OBU initializes a private seed for deriving the vehicle's one-time anonymous secret keys via a reconstructed Key-Der algorithm. Thus, the OBU can avoid pre-storing vast secret keys. In VANETs, the OBU broadcasts its current traffic status to nearby vehicles/RSUs regularly, where the OBU communicates with RSUs and other vehicles via the V2R and V2V respectively.
- *RSU*: This entity acts as a blockchain full node, namely, storing all the transaction data of blockchain. It also provides the APIs for retrieving transactions (defined as *tx_rtv*) and invoking functions of smart contract (including the execution of a message call *eth_calls*). As mentioned above, the RSU responds to the vehicles' requests via the DSRC protocol. In addition, these RSUs are assumed as fully trusted entities and they will not provide false APIs.
- *Blockchain*: This entity is a decentralized ledger maintained by the RSUs, and it is responsible for storing vehicles' public information (e.g., anonymous public keys). The anonymous public keys can be embedded into blockchain as authorized, such that vehicles can check an anonymous public key authorized or not. If yes, the vehicles further verify the authenticity of received messages. Here, any blockchain with the functionality of smart contract (e.g., public Ethereum and permissioned Hyperledger) can be adopted in our EBCPA, only requiring that RSUs act as full nodes to provide the aforementioned services.

3.4 Security and Privacy Requirements

Security and privacy properties are indispensable for a communication protocol to securely share traffic status in VANETs. According to [2], [15], [38], the security and privacy requirements of a BCPPA protocol in VANETs mainly include the following aspects.

- 1) *Message authentication*. Authenticity of transmitted messages (including message integrity and identity legality) should be checked. In other words, any modification on the message will make it invalid. Here, we only consider the message authentication among the V2V and V2R (when sharing traffic status), since the HTTPs protocol can be directly used in the M2B, V2R (during retrieving data from blockchain), and V2C communications.
- 2) *Conditional privacy-preserving*. This property requires that a vehicle's identity privacy can be conditionally hidden. That is, only the manager can obtain the real identity of a vehicle involved in an intercepted message. RSUs and other vehicles can trust the message

received from an authenticated vehicle, but they cannot know its real identity. Hence, the privacy and security of honest vehicles can be efficiently protected, and malicious behaviors (e.g., sharing inaccurate traffic statuses) will be timely tracked and revoked by the manager.

- 3) *Unlinkability*. To prevent the vehicle's driving records from being traced by adversaries, two transmitting messages from the same vehicle are difficult to link.
- 4) *Resilient to birthday collision*. Due to the use of blockchain, a BCPPA protocol should avoid chaining two same blocks (i.e., with the same block hash, but different block contents) simultaneously. That is, it can achieve the birthday collision resilience and hence avoiding disputes among sub-blockchains. Once the BCPPA fails to resist against this attack, the adversary would be able to rewrite transactions and it will lead to an arbitrary authorization of anonymous public keys.
- 5) *Resilient to hijacking*. Also considering the dependency of blockchain, we require that a BCPPA protocol can prevent adversaries from hijacking transactions. That is, all the valid transactions will be chained smoothly. Otherwise, the submission of anonymous public keys will be hindered maliciously, which seriously affects the authorization of vehicles.
- 6) *Resilient to 51% attacks*. A BCPPA protocol should ensure that adversaries are unable to control majority of computing power or consensus ability (i.e., hashrate in PoW and number of permissioned nodes in PBFT [29]). Otherwise, the adversary can reverse and modify history transactions to change the authorization of anonymous public keys.
- 7) *Resilient to other attacks*. A BCPPA protocol in VANETs should also resist against common attacks including impersonation, modification, distributed denial of service, reply, man-in-the-middle, stolen verifier table, and side-channel.

4 OUR PROPOSALS

In this section, we first reconstruct a key derivation scheme for user (e.g., vehicle or RSU) and CA to separately update anonymous secret key and anonymous public key without interactions. Then, a new SoK scheme and smart contract are designed for authentication. On basis of these building blocks, we finally introduce our EBCPA protocol.

4.1 Reconstructed Key Derivation Scheme

In our EBCPA protocol, the manager periodically updates vehicles' anonymous public keys into the smart contract (e.g., pre-submitting about 240 anonymous public keys for an interval of two days at a time [7], [14]). Correspondingly, vehicles can generate fresh anonymous secret keys to authenticate each other. Only those anonymous public keys appeared in smart contract are regarded as authorized. To avoid frequent interactions between vehicles and manager, we reconstruct BIP32² and apply it into our design. Thus, a vehicle interacts with the manager (via M2V) only once for

2. <https://github.com/bitcoin/bips/wiki/Comments:BIP-0032>

sharing its root anonymous public key. Subsequently, the vehicle and manager can independently update the vehicle's new anonymous secret keys and anonymous public keys, respectively. Specifically, the reconstructed KeyDer scheme consists of the following algorithms.

- $(ask_i^{root}, pdi_i^{root}) \leftarrow \text{KGen}(pseed)$. This algorithm is invoked by the i th vehicle to generate the root anonymous secret key ask_i^{root} and root private derivation information pdi_i^{root} . It takes as input a private seed $pseed$ to compute $h_l || h_r = \mathcal{H}(pseed)$, and it sets $ask_i^{root} = h_l \pmod{q}$ and $pdi_i^{root} = h_r \pmod{q}$, then it computes the root anonymous public keys $APK_i^{root,1} = ask_i^{root}P$ and $APK_i^{root,2} = LPK_i + ask_i^{root}LPK_m$. Finally, it returns ask_i^{root} and pdi_i^{root} . Also, the APK_i^{root} and pdi_i^{root} are sent to the manager securely.
- $(ask_i^j, pdi_i^j) \leftarrow \text{PriKD}(ask_i^{j-1}, pdi_i^{j-1}, j)$. This algorithm is invoked by the i th vehicle to derive the j th anonymous secret key. It takes as input the $j-1$ th anonymous secret key ask_i^{j-1} and the $j-1$ th private derivation information pdi_i^{j-1} (or ask_i^{root} and pdi_i^{root} when $j=1$). It first computes $APK_i^{j-1,1} = ask_i^{j-1}P$ and $h_l || h_r = \mathcal{H}(APK_i^{j-1,1} || j, pdi_i^{j-1})$. It finally returns the j th anonymous secret key $ask_i^j = ask_i^{j-1} \cdot h_l \pmod{q}$ and the j th private derivation information $pdi_i^j = h_r \pmod{q}$.
- $(APK_i^{j,1}, APK_i^{j,2}, pdi_i^j) \leftarrow \text{PubKD}(APK_i^{j-1,1}, pdi_i^{j-1}, j, LPK_i, lsk_m)$. This algorithm is invoked by the manager to generate the j th anonymous public key of the i th vehicle. It takes as input the $j-1$ th anonymous public key $APK_i^{j-1,1}$, the $j-1$ th private derivation information pdi_i^{j-1} (or $APK_i^{root,1}$ and pdi_i^{root} when $j=1$), the number j , the long-term public key of vehicle LPK_i and its long-term secret key lsk_m , and computes $h_l || h_r = \mathcal{H}(APK_i^{j-1,1} || j, pdi_i^{j-1})$ as well as setting $APK_i^{j,1} = h_l APK_i^{j-1,1}, pdi_i^j = h_r \pmod{q}$. It also computes $APK_i^{j,2} = LPK_i + lsk_m APK_i^{j,1}$. Finally, it returns $(APK_i^{j,1}, APK_i^{j,2}, pdi_i^j)$.

Security of the above reconstructed KeyDer scheme can be reduced to the discrete logarithm assumption [39]. That is, no $\mathcal{PPT}$ adversary is able to reverse the j th derived anonymous public key $(APK_i^{j,1}, APK_i^{j,2})$ to the root anonymous public key $(APK_i^{root,1}, APK_i^{root,2})$ or previous anonymous public key $(APK_i^{\alpha,1}, APK_i^{\alpha,2})$ (where $\alpha < j$). This property is significant for guaranteeing the anonymity and unlinkability of our EBCPA protocol.

4.2 Designed Signatures of Knowledge Scheme

We design a SoK scheme for vehicles to authenticate each other. The main idea is that the vehicles prove themselves in a zero-knowledge method, namely, owning the corresponding secret key of an anonymous public key without revealing any other information. Also, our SoK scheme works as a signature to authenticate the shared traffic status msg at a specified timestamp ts . Specifically, we denote the above SoK as $SoK\{(lsk_i, ask_i^j, LPK_i) : APK_i^{j,1} = ask_i^j P \wedge APK_i^{j,2} = LPK_i + ask_i^j LPK_m \wedge LPK_i = lsk_i P\}(msg, ts)$, which consists of the following algorithms. Note that, the PASK-BVerfProof is an efficiency-enhancing (batch) verification algorithm for verifying multiple received message/signature pairs at once.

- $(msg_i, \sigma_i, ts_i) \leftarrow \text{PASK-GenProof}(x_i, ask_i^j, lsk_i, msg_i, ts_i)$. This proof generation algorithm takes as input an instance $x_i = (P, APK_i^{j,1}, APK_i^{j,2}, LPK_m)$, $w_i = (ask_i^j, lsk_i, LPK_i)$, candidate message msg_i and a timestamp ts_i . It first randomly chooses $r_{i,1}, r_{i,2} \in \mathbb{Z}_q^*$ to compute $R_{i,1} = r_{i,1}P$, $R_{i,2} = r_{i,2}P + r_{i,1}LPK_m$, $c_i = \mathcal{H}(x_i || msg_i || ts_i || P || APK_i^{j,1} || APK_i^{j,2} || LPK_m || R_{i,1} || R_{i,2})$. Then, it computes $z_{i,1} = r_{i,1} - ask_i^j \cdot c_i \pmod{q}$, $z_{i,2} = r_{i,2} - lsk_i \cdot c_i \pmod{q}$. Finally, it returns $(msg_i, \sigma_i = (c_i, z_{i,1}, z_{i,2}, R_{i,1}, R_{i,2}), ts_i)$.
- $\{0, 1\} \leftarrow \text{PASK-VerfProof}(x_i, msg_i, \sigma_i, ts_i)$. This verification algorithm takes as input the instance $x_i = (P, APK_i^{j,1}, APK_i^{j,2}, LPK_m)$, a candidate message/signature pair (msg_i, σ_i) and a timestamp ts_i . It first parses $\sigma_i = (c_i, z_{i,1}, z_{i,2}, R_{i,1}, R_{i,2})$, and then computes $c_i = \mathcal{H}(x_i || msg_i || ts_i || P || APK_i^{j,1} || APK_i^{j,2} || LPK_m || R_{i,1} || R_{i,2})$. Finally, it checks if $R_{i,1} + R_{i,2} = (z_{i,1} + z_{i,2})P + z_{i,1}LPK_m + c_i(APK_i^{j,1} + APK_i^{j,2})$ holds or not. If yes, it returns 1 to represent the validity of (msg_i, σ_i, ts_i) , and 0 otherwise.
- $\{0, 1\} \leftarrow \text{PASK-BVerfProof}(\{x_i, msg_i, \sigma_i, ts_i\}_{i=1}^n)$. This batch verification algorithm takes as input the n -tuple instances, candidate message/signature pairs and timestamps $\{x_i, msg_i, \sigma_i, ts_i\}_{i=1}^n$. It first parses $x_i = (P, APK_i^{j,1}, APK_i^{j,2}, LPK_m)$, $\sigma_i = (c_i, z_{i,1}, z_{i,2}, R_{i,1}, R_{i,2})$, $\forall i \in \{1, 2, \dots, n\}$. Then, it chooses a random vector $\vec{v} = \{v_1, v_2, \dots, v_n\}$, where $v_j \in [1, 2^t], \forall i \in \{1, 2, \dots, n\}$ and t is a small integer (with little computation costs). It is worth noting that, if without these random integers, an adversary can inject two invalid message/signature pairs which are negatives with each other. As a result, the verifier will accept the whole batching message/signature pairs and timestamps (including the two invalid ones).

Next, it computes $c_i = \mathcal{H}(x_i || msg_i || ts_i || P || APK_i^{j,1} || APK_i^{j,2} || LPK_m || R_{i,1} || R_{i,2})$, $\forall i \in \{1, 2, \dots, n\}$. Finally, it checks if $\sum_{i=1}^n [v_i(R_{i,1} + R_{i,2})] = [\sum_{i=1}^n v_i(z_{i,1} + z_{i,2})]P + (\sum_{i=1}^n v_i z_{i,1})LPK_m + \sum_{i=1}^n [v_i c_i (APK_i^{j,1} + APK_i^{j,2})]$. It returns 1 if this equation hold, and 0 otherwise.

Our proposed SoK scheme can satisfy the security properties of completeness, soundness, and zero-knowledge. The concrete proof will be given in Section 5.

4.3 Design of Smart Contract

Smart contract generally refers to automation of legal contracts, and now it widely represents code scripts running synchronously in blockchain [40]. In a blockchain system (e.g., Ethereum, Hyperledger), each contract is with a unique address and their equipping algorithms can provide various services (e.g., anonymous public key management in this paper). One can invoke the algorithms via submitting the smart contract address and corresponding parameters (e.g., algorithm name and arguments).

In our designed EBCPA protocol, we use smart contract to manage anonymous public keys. One anonymous public key is authorized if and only if it was submitted into the smart contract, and the submission operation can be only executed by the manager. Then, the vehicles can authenticate each other via acting as the submitted anonymous public key. Additionally, the manager can revoke all the related

anonymous public keys when he/she finds a vehicle broadcasting counterfeit messages maliciously. Thus, our designed smart contract mainly includes three algorithms, namely, Submit, Check and Revoke (shown in Algorithm 1). It is worth noting that our smart contract can be deployed in public Ethereum or permissioned Hyperledger according to different goals such as robustness or efficiency.

Algorithm 1. Smart Contract on *APKList*

Require: Function name, invoked parameters

Ensure: *Setting up functions:*

 address *manager*; % The address of manager.

function *APKList*()

 % Constructor, automatically executed during the deployment.

manager = *msg.sender*; % Initialize the issuer as manager.

struct *APK*

*APK*₁; % uint256[2], one part of an APK.

*APK*₂; % uint256[2], the other part of an APK.

mapping (uint256 => *APK*) *apk* % A mapping of APK.

function *Submit*(uint256[2] *APK*₁, uint256[2] *APK*₂) *public* returns (address *addr*)

 % Invoked by manager to add a new anonymous public key.

 require(*msg.sender* == *manager*); % Only the manager can successfully invoke this algorithm.

index = $\mathcal{H}(\text{APK}_1, \text{APK}_2)$; % Compute the index for subsequently retrieving anonymous public keys.

apk[*index*].*APK*₁ = *APK*₁;

apk[*index*].*APK*₂ = *APK*₂;

return *msg.sender*;

function *Check*(uint256[2] *APK*₁, uint256[2] *APK*₂) *view* returns (bool)

 % Invoked by anyone to check if one APK is authorized or not.

index = $\mathcal{H}(\text{APK}_1, \text{APK}_2)$;

if *apk*[*index*].*APK*₁ = *APK*₁ $\wedge$ *apk*[*index*].*APK*₂ = *APK*₂ **then** **return**1;

return0;

function *Revoke*(uint256[2] *APK*₁, uint256[2] *APK*₂) *public* returns (bool)

 % Invoked by manager to revoke an APK.

 require(*msg.sender* == *manager*); % Only the manager can successfully execute the revocation.

index = $\mathcal{H}(\text{APK}_1, \text{APK}_2)$;

if *apk*[*index*].*APK*₁ = *APK*₁ $\wedge$ *apk*[*index*].*APK*₂ = *APK*₂ **then** *apk*[*index*].*APK*₁ = NULL;

apk[*index*].*APK*₂ = NULL;

return1;

return0;

4.4 Constructed EBCPA Protocol

On basis of the above system building blocks (i.e., KeyDer scheme, SoK scheme and smart contract), we will introduce the construction of EBCPA in this section.

4.4.1 System Setup Phase

This phase is performed by vehicles, CA and manager to register/update the real/anonymous identities of vehicles. It consists of the following steps.

- 1) Each vehicle randomly chooses $lsk_i \in \mathbb{Z}_q^*$ as its long-term secret key and computes $LPK_i = lsk_i P$ as the

long-term public key. Then, it applies for the certificate $cert_i$ of LPK_i from the CA via submitting its identification information online/offline. After checking the validity of application information, the CA computes and returns the certificate $cert_i$ to the vehicle. Note that we suggest using the blockchain-based certificate management mechanism [38] for reducing the complexity, and certainly the existing public key certificate system (e.g., X.509) can be also adopted directly.

- 2) To register the anonymous public key from manager, the vehicle first randomly chooses a private seed and generates the secret information (i.e., a root anonymous secret key ask_i^{root} and root private derivation information pdi_i^{root}). Notably, the ask_i^{root} and pdi_i^{root} will be securely stored in the vehicle's OBU and then updated periodically. For example, deriving about 240 times for an interval of two days at a time [7], [14]. It also computes the root anonymous public keys $APK_i^{root,1} = ask_i^{root} P$ and $APK_i^{root,2} = LPK_i + ask_i^{root} LPK_m$. Then, the vehicle uses its long-term secret key lsk_i to sign $(APK_i^{root,1}, APK_i^{root,2}, pdi_i^{root})$ via ESign, that is, $S_i = \text{ESign}(lsk_i, APK_i^{root,1}, APK_i^{root,2}, pdi_i^{root})$. Finally, the vehicle sends $(APK_i^{root,1}, APK_i^{root,2}, pdi_i^{root}, S_i, cert_i)$ to the manager via a secure channel.

The manager uses the $cert_i$ (probably retrieved from blockchain) to check the authenticity of LPK_i , and then he/she invokes $\text{EVerif}(LPK_i, APK_i^{root,1}, APK_i^{root,2}, pdi_i^{root}, S_i)$ to check if $(APK_i^{root,1}, APK_i^{root,2}, pdi_i^{root})$ is actually from the vehicle of long-term public key LPK_i .

- 3) The manager deploys the smart contract *APKList* into the blockchain and he/she will obtain a smart contract identity *sid*. The *sid* will be public for all the vehicles and RSUs to trigger this smart contract subsequently. Note that this deployment is executed only once.

At the first time of submitting the anonymous public key into the *sid*, the manager uses its long-term secret key lsk_m to check if $APK_i^{root,2} = LPK_i + lsk_m APK_i^{root,1}$. If not, this request will be discarded; Otherwise, the manager invokes *Submit* to submit the anonymous public key $(APK_i^{root,1}, APK_i^{root,2})$ into the *sid*. For submitting the *j*th (where $j > 1$) anonymous public key, the manager needs to invoke $(APK_i^{j,1}, APK_i^{j,2}, pdi_i^j) \leftarrow \text{PubKD}(APK_i^{j-1,1}, j, LPK_i, lsk_m)$ prior to submitting $(APK_i^{j,1}, APK_i^{j,2})$. In our EBCPA protocol, the manager periodically submits a new $(APK_i^{j,1}, APK_i^{j,2})$ into the *sid* such that vehicles can anonymously share their traffic statuses.

4.4.2 Message Signing Phase

This phase is performed by vehicles to compute a message/signature pair for authenticating their identities and sharing messages. In VANETs, the generated message/signature pair will be broadcast to nearby RSUs and vehicles through a wireless communication method (e.g., DSRC). For a clearer description, we suppose that vehicle V_a would like to broadcast a message msg_a to nearby vehicles (e.g., V_b) or RSUs, it will proceed the following steps.

- 1) Assuming that the V_a 's current anonymous secret key is ask_a^{k-1} and current private derivation information is pdi_a^{k-1} , then it invokes $(ask_a^k, pdi_a^k) \leftarrow \text{PriKD}(ask_a^{k-1}, pdi_a^{k-1}, k)$ and computes $APK_a^{k,1} = ask_a^k P$, $APK_a^{k,2} = LPK_a + ask_a^k LPK_m$, where LPK_a is the long-term public key of vehicle V_a and LPK_m is the long-term public key of manager. This process can be orderly precomputed several times before each communication, such that the vehicle can directly adopt a new anonymous public key to share its message.
- 2) The vehicle V_a invokes $(msg_a, \sigma_a, ts_a) \leftarrow \text{PASK-GenProof}(x_a, ask_a^k, lsk_a, msg_a, ts_a)$ to generate the signature of msg_a , where $x_a = (P, APK_a^{k,1}, APK_a^{k,2}, LPK_m)$, ts_a is the current timestamp. The (msg_a, σ_a, ts_a) will be sent to the vehicle V_b . Also, the $(APK_a^{k,1}, APK_a^{k,2})$ in x_a needs to be sent each time for verification.

4.4.3 Message Verification Phase

This phase is executed by a vehicle or RSU to validate a received message/signature pair and timestamp. The content of message (e.g., traffic status) will be accepted if and only if the message/signature pair and timestamp pass the verification. To verify a single message/signature pair and timestamp, namely, suppose the V_b receives (msg_a, σ_a, ts_a) and $(APK_a^{k,1}, APK_a^{k,2})$ from the vehicle V_a . Then, the vehicle V_b directly invokes $\text{Check}(APK_a^{k,1}, APK_a^{k,2})$ in smart contract *scid* to check if the anonymous public key $(APK_a^{k,1}, APK_a^{k,2})$ is revoked or not. Here, the smart contract *scid* was deployed in the blockchain during the setup phase. If not, V_b continues to invoke PASK-VerfProof to check the validity of (msg_a, σ_a, ts_a) . If invalid, V_b discards this traffic status; Otherwise, V_b believes the message msg_a is valid (i.e., msg_a is indeed received from the authentic V_a).

In addition to the above single verification of one message, we also provide a batch verification of multiple messages for improving the verification efficiency. Assuming that the vehicle V_b has received n -tuple message/signature pairs, timestamps $(msg_i, \sigma_i, ts_i)_{i=1}^n$ and one-time anonymous public keys $(APK_i^{j,1}, APK_i^{j,2})_{i=1}^n$ from different vehicles. The V_b first invokes $\text{Check}(APK_i^{j,1}, APK_i^{j,2}), \forall i \in [1, 2, \dots, n]$ from *scid* to check if the anonymous public keys are revoked or not. If not, V_b continues to invoke PASK-BVerfProof to check the validity of $(msg_i, \sigma_i, ts_i)_{i=1}^n$. If valid, V_b accepts these messages; Otherwise, it rejects.

5 SECURITY ANALYSIS

This section demonstrates that our EBCPA protocol can satisfy the aforementioned security requirements, which is mainly owing to the security of designed SoK scheme. Before analyzing the security of EBCPA, we first prove the security of designed SoK scheme as follows.

Theorem 1. *The above SoK scheme is a non-interactive zero-knowledge argument in the random oracle model. That is, it satisfies Perfect completeness, Soundness and Zero-knowledge.*

Proof. We prove the (perfect) completeness, soundness, and zero-knowledge of our SoK scheme as follows.

(Perfect) Completeness. This property represents two consistencies, namely, between PASK-GenProof and PASK-VerfProof , PASK-GenProof and PASK-BVerfProof . The former

refers to that a message/signature pair and timestamp generated by PASK-GenProof will be accepted by PASK-VerfProof , and the latter refers to multiple message/signature pairs and timestamps generated by PASK-GenProof will be accepted by PASK-BVerfProof at once. The proposed SoK scheme satisfies this property for the following equations always hold.

$$\begin{aligned}
 c_i &= \mathcal{H}(x_i || msg_i || ts_i || P || APK_i^{j,1} || APK_i^{j,2} || LPK_m || R_{i,1} || R_{i,2}), \\
 R_{i,1} + R_{i,2} &= (z_{i,1} + z_{i,2})P + z_{i,1}LPK_m + c_i(APK_i^{j,1} + APK_i^{j,2}) \\
 &= r_{i,1}P - ask_i^j \cdot c_i P + r_{i,2}P - lsk_i \cdot c_i P + r_{i,1}LPK_m \\
 &\quad - ask_i^j \cdot c_i LPK_m + c_i(ask_i^j P + LPK_i + ask_i LPK_m) \\
 &= (r_{i,1} + r_{i,2})P + r_{i,1}LPK_m - c_i ask_i P - c_i LPK_i \\
 &\quad - c_i ask_i^j LPK_m + c_i ask_i^j P + c_i LPK_i + c_i ask_i LPK_m \\
 &= R_{i,1} + R_{i,2},
 \end{aligned}$$

$$\forall i \in [1, \dots, n],$$

$$\begin{aligned}
 c_i &= \mathcal{H}(x_i || msg_i || ts_i || P || APK_i^{j,1} || APK_i^{j,2} || LPK_m || R_{i,1} || R_{i,2}), \\
 \Sigma_{i=1}^n [v_i(R_{i,1} + R_{i,2})] &= \Sigma_{i=1}^n v_i(z_{i,1}P + c_i APK_i^{j,1} + z_{i,2}P + z_{i,1}LPK_m \\
 &\quad + c_i APK_i^{j,2}) = \Sigma_{i=1}^n v_i(r_{i,1} - ask_i^j \cdot c_i)P \\
 &\quad + \Sigma_{i=1}^n (v_i c_i ask_i^j P) + \Sigma_{i=1}^n v_i(r_{i,2} - lsk_i \cdot c_i)P \\
 &\quad + \Sigma_{i=1}^n v_i(r_{i,1} - ask_i^j \cdot c_i)LPK_m \\
 &\quad + \Sigma_{i=1}^n v_i c_i (LPK_i + ask_i LPK_m) \\
 &= \Sigma_{i=1}^n v_i r_{i,1}P + \Sigma_{i=1}^n v_i r_{i,2}P + \Sigma_{i=1}^n v_i r_{i,1}LPK_m \\
 &= \Sigma_{i=1}^n (v_i(R_{i,1} + R_{i,2})).
 \end{aligned}$$

Soundness. The soundness follows from the property of special soundness of Σ -protocol, but simulating the $\mathcal{H}$ as a random oracle instead. Suppose a $\mathcal{PPT}$ prover P^* generates an accepted instance and message/signature pair (x_i, msg_i, σ_i) and timestamp ts_i , where $\sigma_i = (c_i, z_{i,1}, z_{i,2}, R_{i,1}, R_{i,2})$. Then, we construct such an extractor EXT, where upon seeing the above accepted pair, EXT rewinds P^* to the oracle $\mathcal{H}(x_i || msg_i || ts_i || P || APK_i^{j,1} || APK_i^{j,2} || LPK_m || R_{i,1} || R_{i,2})$ that returns c_i . It then reprograms the random oracle such that $\hat{c}_i = \mathcal{H}(x_i || msg_i || ts_i || P || APK_i^{j,1} || APK_i^{j,2} || LPK_m || R_{i,1} || R_{i,2})$ and continues the rest execution of P^* . In an expected polynomial time, another valid message/signature pair and timestamp will be generated, namely, $(x_i, msg_i, \hat{\sigma}_i, ts_i)$, where $\hat{\sigma}_i = (\hat{c}_i, \hat{z}_{i,1}, \hat{z}_{i,2}, R_{i,1}, R_{i,2})$. According to the equations $z_{i,1} = r_{i,1} - ask_i^j \cdot c_i \pmod{q}$, $z_{i,2} = r_{i,2} - lsk_i \cdot c_i \pmod{q}$ and $\hat{z}_{i,1} = r_{i,1} - ask_i^j \cdot \hat{c}_i \pmod{q}$, $\hat{z}_{i,2} = r_{i,2} - lsk_i \cdot \hat{c}_i \pmod{q}$, the witnesses can be extracted via computing the following equations.

$$\begin{aligned}
 ask_i^j &= (z_{i,1} - \hat{z}_{i,1})(\hat{c}_i - c_i)^{-1} \pmod{q}, \\
 lsk_i &= (z_{i,2} - \hat{z}_{i,2})(\hat{c}_i - c_i)^{-1} \pmod{q}.
 \end{aligned}$$

Note that $\hat{c}_i - c_i$ is prime to q since both $\hat{c}_i$ and c_i lie in $\mathbb{Z}_q^*$ and q is a large prime as mentioned. This means that $(\hat{c}_i - c_i)^{-1} \pmod{q}$ can be efficiently computed by the extended euclidean algorithm.

Zero-Knowledge. To prove this property, we construct a simulator Sim (see Algorithm 2) that simulates all interactions with any verifier V^* except the setting that Sim controls hash function $\mathcal{H}$ as a random oracle $\mathcal{O}_\mathcal{H}$. Finally, it generates the message/signature pair and timestamp $(x_i, \text{msg}_i, \sigma_i = (c_i^*, z_{i,1}^*, z_{i,2}^*, R_{i,1}^*, R_{i,2}^*), ts_i)$.

Observing that this message/signature pair and timestamp will be accepted by the V^* because the secure hash function is controlled as a random oracle. In addition, the $c_i^*, z_{i,1}^*, z_{i,2}^*$ in Sim have the same distribution as that in real protocol execution. Also, the $R_{i,1}^*$ and $R_{i,2}^*$ are uniquely determined by $c_i^*, z_{i,1}^*, z_{i,2}^*$, according to the computations $R_{i,1}^* = z_{i,1}^* P + c_i^* \text{APK}_i^{j,1}$ and $R_{i,2}^* = z_{i,2}^* P + z_{i,1}^* \text{LPK}_m + c_i^* \text{APK}_i^{j,2}$. This means that the distribution of $R_{i,1}^*$ and $R_{i,2}^*$ is also identical to that in the real protocol execution, and hence the transcript $(x_i, \text{msg}_i, \sigma_i, ts_i)$ output by the Sim has the same distribution as that output by an honestly generated transcript. Therefore, we can conclude that our proposed SoK scheme satisfies the zero-knowledge, for the Sim does not involve any information of witness $w_i = (lsk_i, ask_i^j, \text{LPK}_i)$. $\square$

Then, we characterize the anonymity property of the EBCPA protocol, which ensures that traffic data reveal no real identity information of RSU or vehicles to the adversary. That is, no $\mathcal{PPT}$ adversary $\mathcal{A}$ can distinguish between two message/signature pairs, where the long-term public keys are chosen by $\mathcal{A}$. To achieve a more clearer and formal definition, we denote $(\text{LPK}, lsk) \leftarrow \text{LKG}(pp)(\lambda)$ as the vehicle's long-term public/secret key generation algorithm, $(\text{APK}, ask, pdi^i) \leftarrow \text{AKG}(lsk, pdi^{i-1}, \text{LPK}_m, i)$ as the vehicle's anonymous public/secret key generation algorithm, $(m, \sigma, ts) \leftarrow \text{Sign}(ask, m, ts)$ as the message signing algorithm, and $\{0, 1\} \leftarrow \text{Verify}(\text{APK}, m, \sigma, ts)$ as the verification algorithm.

Algorithm 2. Simulator Sim

Require: a security parameter λ .

Ensure: the message/signature pair (msg_i, σ_i) and timestamp ts_i .

- 1: Given an instance $x_i = (P, \text{APK}_0^{j,1}, \text{APK}_0^{j,2}, \text{LPK}_m)$ to be proved.
 - 2: Randomly choose $c_i^*, z_{i,1}^*, z_{i,2}^* \in \mathbb{Z}_q^*$.
 - 3: Compute $R_{i,1}^* = z_{i,1}^* P + c_i^* \text{APK}_i^{j,1}$ and $R_{i,2}^* = z_{i,2}^* P + z_{i,1}^* \text{LPK}_m + c_i^* \text{APK}_i^{j,2}$.
 - 4: Set $\mathcal{O}_\mathcal{H}(x_i || \text{msg}_i || ts_i || P || \text{APK}_i^{j,1} || \text{APK}_i^{j,2} || \text{LPK}_i || \text{LPK}_m || R_{i,1}^* || R_{i,2}^*) = c_i^*$.
 - 5: **return** $(\text{msg}_i, \sigma_i = (c_i^*, z_{i,1}^*, z_{i,2}^*, R_{i,1}^*, R_{i,2}^*), ts_i)$.
-

Definition 1 (Anonymity). A EBCPA protocol $\Pi = (\text{LKG}, \text{AKG}, \text{Sign}, \text{Verify})$ owns the anonymity if no $\mathcal{PPT}$ adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ can distinguish the following experiments with a non-negligible probability.

$$\begin{aligned} & \text{EXP}_{\text{an}}^b(\Pi, \mathcal{A}, \lambda) : \\ & (\text{LPK}_m, lsk_m) \leftarrow \text{LKG}(\lambda), \\ & ((\text{LPK}_0, lsk_0), (\text{LPK}_1, lsk_1)) \leftarrow \mathcal{A}_1^{\text{LKG}(\lambda)}, \\ & (\text{APK}_b, ask_b, pdi^i) \leftarrow \text{AKG}(lsk, pdi^{i-1}, \text{LPK}_m, i), \\ & (m, \sigma_b, ts) \leftarrow \text{Sign}(ask_b, m, ts), \\ & b' \leftarrow \mathcal{A}_2(m, \sigma_b, ts). \end{aligned}$$

That is,

$$|\Pr[\text{EXP}_{\text{an}}^0(\Pi, \mathcal{A}, \lambda) = 1] - \Pr[\text{EXP}_{\text{an}}^1(\Pi, \mathcal{A}, \lambda) = 1]| \leq \text{negl}(\lambda).$$

Theorem 2. The EBCPA protocol $\Pi = (\text{LKG}, \text{AKG}, \text{Sign}, \text{Verify})$ satisfies the property of anonymity iff the adopted SoK scheme owns the zero-knowledge property.

Proof. To prove the anonymity property, we present the following hybrid experiments $(\text{EXP}_{\text{an}}^0, E_1, E_2, \text{EXP}_{\text{an}}^1)$.

- **Experiment E_1 .** E_1 modifies EXP_{an}^0 by adopting the simulator $\mathcal{S}$ (see Algorithm 2) to compute the signature when executing the Sign algorithm. Specifically, the challenger simulates an instance $x_0 = (P, \text{APK}_0^{j,1}, \text{APK}_0^{j,2}, \text{LPK}_m)$ and computes $(m, \sigma_0, ts) \leftarrow \mathcal{S}(1^\lambda)$ but not the PASK-GenProof algorithm when executing the Sign. Due to the zero-knowledge property of adopted SoK scheme, and the E_1 owns identical message/signature pair distribution with EXP_{an}^0 ; hence, we have

$$|\Pr[\text{EXP}_{\text{an}}^0(\Pi, \mathcal{A}, \lambda) = 1] - \Pr[E_1(\Pi, \mathcal{A}, \lambda) = 1]| \leq \text{negl}(\lambda).$$

- **Experiment E_2 .** E_2 modifies E_1 when invoking the Sign algorithm. The challenger simulates the other instance $x_1 = (P, \text{APK}_1^{j,1}, \text{APK}_1^{j,2}, \text{LPK}_m)$, which is the same as that in EXP_{an}^1 and computes $(m, \sigma_1, ts) \leftarrow \mathcal{S}(1^\lambda)$.

We also have $\{x_0, m, \sigma_0, ts\} \stackrel{c}{\approx} \{x_1, m, \sigma_1, ts\}$ due to the zero-knowledge property of proposed SoK scheme. Hence, the following is satisfied.

$$|\Pr[E_1(\Pi, \mathcal{A}, \lambda) = 1] - \Pr[E_2(\Pi, \mathcal{A}, \lambda) = 1]| \leq \text{negl}(\lambda).$$

The zero-knowledge property also implies that

$$|\Pr[E_2(\Pi, \mathcal{A}, \lambda) = 1] - \Pr[\text{EXP}_{\text{an}}^1(\Pi, \mathcal{A}, \lambda) = 1]| \leq \text{negl}(\lambda).$$

As discussed above, we can conclude the entire probability that the adversary $\mathcal{A}$ distinguishes the hybrid experiments as follows.

$$|\Pr[\text{EXP}_{\text{an}}^0(\Pi, \mathcal{A}, \lambda) = 1] - \Pr[\text{EXP}_{\text{an}}^1(\Pi, \mathcal{A}, \lambda) = 1]| \leq \text{negl}(\lambda). \quad \square$$

On basis of the above theorems, we can further discuss how our proposal meets the aforementioned security and privacy requirements.

Message Authentication. According to the perfect completeness and soundness of adopted SoK scheme, no $\mathcal{PPT}$ adversary can forge a valid message/signature pair without the anonymous or long-term secret keys. Moreover, the chained anonymous public key in smart contract can be used to verify the authenticity of sender. Thus, the receiver confirms the authenticity of a receiving message/signature pair $(x_a, \text{msg}_a,$

3. The notation $X \stackrel{c}{\approx} Y$ represents that two distribution ensembles X and Y are computationally indistinguishable.

σ_a) and timestamp ts_a via checking if $\text{PASK-VerProof}(x_a, msg_a, \sigma_a, ts_a) = 1$ and $\text{Check}(APK_a^1, APK_a^2) = 1$ hold. The former equation is to validate a message/signature pair and timestamp, and the latter is to ensure that the sender is indeed authorized.

Conditional Privacy-Preserving. Our proposal adopts a fresh anonymous public/secret key pair in each authentication. On the one hand, the aforementioned anonymity of EBCPA implies that the anonymous public key can efficiently hide the information of long-term public key, that is, the *PPT* receiver and adversaries only obtain the knowledge of authorized anonymous public keys but not the real identities. On the other hand, the manager who owns the tracing key lsk_m can track the real identity (i.e., long-term public key). Specifically, to trace an anonymous key $(APK_i^{j,1}, APK_i^{j,2})$, the manager uses lsk_m to compute the long-term public key via $LPK_i = APK_i^{j,2} - lsk_m APK_i^{j,1}$.

Unlinkability. As mentioned above, each authentication in our proposal involves a fresh anonymous public/secret key pair. This pair is derived from a root/previous anonymous public/secret key pair and private derivation information based on the KeyDer algorithm. Thus, to link two different messages to the same sender, one should be able to derive one anonymous public key to the other. However, the derivation algorithm of anonymous public/secret key pairs requires a private derivation information which is only known by the sender itself. In addition, the anonymity of EBCPA further prevents the adversary from distinguishing two different message/signature pairs. Thus, our proposal can satisfy this security requirement.

Resilient to Birthday Collision. Our proposal can satisfy this property if the adopted blockchain can resilient to birthday collision. We use the mature blockchain systems (e.g., Ethereum and Hyperledger) supporting smart contract for our design. These blockchain systems adopt secure hash functions such as Keccak256 in [41] and SHA256 [42] to compute the block hash, and hence they can efficiently avoid generating two blocks of birthday collision.

Resilient to Hijacking. All the transactions are comprised of a digital signature scheme (e.g., ECDSA) and broadcast in the blockchain network. This means that those valid transactions cannot be tampered by *PPT* adversaries successfully, and they will be recored into the blockchain (via different recording nodes) sooner or later once valid.

Resilient to 51% Attacks. This property is also due to the adopted blockchain system (i.e., Ethereum and Hyperledger). In popular blockchain systems, the main method to resist this attack is to higher the execution cost. For instance, Ethereum uses a PoW with "ASIC-resistant" consensus mechanism to reduce economic incentives of centralized mining. Hyperledger applies a permissioned paradigm, namely, assuming that the recording nodes are all trusted and difficult to be corrupted.

Resilient to Other Attacks. We also demonstrate that our proposal can resist the following common attacks.

- **Impersonation.** Once the adversary would like to impersonate an authorized vehicle (e.g., Alice), it needs to compute a valid signature as Alice for a targeted message. However, it is impossible for a *PPT* adversary due to the message authentication property, that is,

the verifier can easily detect this attack via verifying the receiving message/signature pair and timestamp.

- **Modification.** We suppose that a *PPT* adversary tempts a transmitting message msg' , it will be found and leads to the msg' being discarded by the receiver. Due to the soundness of SoK, the adversary is unable to forge a valid signature for msg' without the knowledge of receiver's anonymous or long-term secret key. Thus, any modification on msg' will make the verification of original signature false.
- **DDoS.** This property also benefits from adopted blockchain systems (e.g., Ethereum and Hyperledger). In Ethereum, the execution of DDoS attacks will involve economically expensive costs (including transaction fees and gas costs). This huge cost is really the advantage of Ethereum to resist DDoS attacks, by comparing to the regular Internet where a server responds to user's request for free. In Hyperledger, it does not require any transaction fee, but the trusted permissioned nodes can adaptively discard those malicious transactions and hence Hyperledger can resist DDoS attacks.
- **Replay.** In each authentication, the vehicle derives a fresh anonymous public/secret key pair. Moreover, the transmitting message/signature pair is embedded with a current timestamp. These can effectively keep the freshness of transmitting message and hence promote the detection of replay attacks timely.
- **Man-in-the-middle.** According to the aforementioned discussion of message authentication, our proposal can provide secure mutual authentication among vehicles.
- **Stolen verifier table.** This attack represents that an adversary can steal or tamper passwords or verification tables stored in a server's database. In the whole authentication process, our proposal only involves the proposed SoK without the need to maintain a verifier table. The only stored anonymous public keys in blockchain are public and immutable due to the immutability of blockchain. Thus, the adversary cannot steal any verifier table for launching other attacks.
- **Side-channel.** Our proposal only stores the current anonymous secret key and private derivation information in the OBU. That is, these secrets will be updated periodically (i.e., our adopted OBU is realistic but not trustworthy), meaning that it is more difficult to launch success side-channel attacks on our proposal than that on existing ID-based solutions (e.g., [10], [15]). Indeed, embedding several secrets into the OBUs is common in most existing authentication protocols. Therefore, in terms of resilience to side-channel attacks, our proposal and these protocols have the similar security level. To further enhance the protection of these secrets and resist against powerful side-channel attacks, multiplicative secret sharing technique [16] can be adopted.

6 PERFORMANCE EVALUATION

To show the utility of our EBCPA protocol, we first implement it through the prototype of blockchain in Ethereum

TABLE 2
Notations, Descriptions and Size

Notation	Description	Size (byte)	Notation	Description
$ \mathbb{G} $	Size of an element in group $\mathbb{G}$	64	T_{sm}	Time of scale multiplication in $\mathbb{G}$
$ \mathbb{Z}_q^* $	Size of an element in field $\mathbb{Z}_q^*$	32	T_{me}	Time of modular exponentiation in $\mathbb{Z}_q^*$
$ Tid $	Size of a transaction hash in Ethereum	32	T_{pa}	Time of point addition in $\mathbb{G}$
$ msg $	Size of a message in our simulation	32	T_{mi}	Time of modular inversion in $\mathbb{Z}_q^*$
$ ts $	Size of a timestamp in our simulation	8	T_{mm}	Time of modular multiplication in $\mathbb{Z}_q^*$
$ ID $	Size of an entity identity	8	T_h	Time of hash function <i>keccak256</i>
$ c_{ABE} $	Size of a ABE ciphertext	$128l + 64$	T_{AEnc}	Time of ABE encryption algorithm
			T_{ADec}	Time of ABE decryption algorithm
			T_{enc}	Time of AES encryption algorithm
			T_{dec}	Time of AES decryption algorithm

*The attribute-based encryption (ABE) scheme is directly cited from [20], among which l is the number of leaf nodes in the access tree.

and Hyperledger test networks, vehicle authentication efficiency and simulated VANETs scenario. Then, we discuss the efficiency according to the implementation results.

6.1 Implementation

To implement the blockchain-enabled prototype, we first adopt an on-line public Ethereum test network (i.e., Rinkeby⁴) to deploy our design smart contract, where Rinkeby provides IDE (Integrated Development Environment) for compiling and deploying Solidity smart contract conveniently. Specifically, our adopted configuration of Remix is compiler (0.4.19+commit.c4cbbb05), language (Solidity), EVM version (compiler default), Deployment Environment (JavaScript Virtual Machine) and Featured Plugins (Debugger, Deploy and Run Transactions, Solidity Compiler, Solidity Static Analysis, and Solidity Uint Testing). After compiling our written Solidity smart contract code, we deploy it into the above configured Remix to measure the gas cost of each algorithm. Here, the gas cost is one of concerned factors in Ethereum, which reflects the cost of using Ethereum typological blockchain [43].

We also implement our prototype based on the permissioned Hyperledger to further show its compatibility. Concretely, we build a test network comprising two organizations and each organization is with one peer node. This simulation platform is with the operation system (Ubuntu TLS 19.0.4), CPU (Inter(R) Core(TM) i7-9750H, 2.60GHz), and RAM (1.9 GB), Hyperledger (V2.1) and Caliper (V0.4.0). Involved dependencies include docker (V19.03.6), docker-compose (V2.0.1), go (V1.17), Node.js (V10.22.1), npm (V6.14.6), and npx (V10.2.2). In the test network, we create a channel named as *mychannel* and instantiate our chaincode (i.e., *apklist*) in it. For accuracy, we run 500 test times on random samples for every algorithm in the chaincode, and finally we obtain the latency and throughput.

In addition, we use our personal computer (PC) to theoretically and practically evaluate the authentication efficiency in terms of communication and computation costs. In the theoretical analysis, involved notations and corresponding size are listed in Table 2. In the practical evaluation, the PC is configured with operation system (Windows 10, 64 bits), CPU (Inter(R) Core(TM) i7-9750H, 2.60GHz) and RAM (16.0 GB), the cryptographic library is Miracl

V7.0, and the elliptic curve is Barreto-Naehrig (BN) curve over base field $\mathbb{F}_{256}$.

Finally, to discuss the average message authentication delay and average message loss rate of the EBCPA, we simulate it via VanetMobiSim⁵ and NS-2⁶ in the simulation environment of VANETs. The concrete configuration of PC is with operation system (Ubuntu 16.04, 64 bits), CPU (Intel(R) Core(TM) i7-9750H, 2.60GHz), and RAM (4GB). As shown in Fig. 3, our simulated scenario is performed in a map with four $0.5 \times 0.5 \text{ km}^2$ blocks and each block is equipped with a RSU (whose communication range is 600 miles). Vehicles have different average speeds (i.e., ranging from 5 miles/second to 70 miles/second) and the same communication range (i.e., 300 miles/second), and their broadcast message interval, broadcast bandwidth bound, and packet size are set as 100 milliseconds, 6 Mbps and 264 bytes, respectively. Other parameters like Channel, Propagation, Phy, Mac, Queue, and Antenna are chosen as WirelessChannel, TwoRay-Ground, WirelessPhy, 802_11, DropTail/PriQueue, and OmniAntenna, respectively. The running time of each simulation is 100 seconds. Interested readers could refer to our source codes⁷ for more details of the above implementations.

6.2 Benchmarks

Gas cost. From the above blockchain-enabled prototype in Ethereum, we can obtain the involved gas cost in performing our proposal. Specifically, the Gas Limit is set as 10,000,000gas and each gas is worth 2GWei (corresponding to 0.006Ether or 1.35USD for an exchange rate 225USD/Ether). As shown in Fig. 4, most gas cost is the deployment of smart contract with 0.5414USD transaction cost (which includes the 0.2229USD execution cost). The costs of executing Submit, Check, and Revoke are all less than 0.1USD, especially the Check only requires 0.0159USD (including 0.0012USD). Deploying smart contract is performed only once, and the Check will be invoked in each verification, and thus such a cost is acceptable.

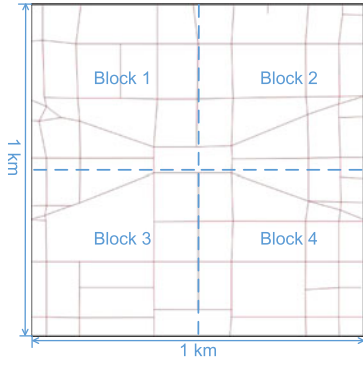
Latency and throughput. On basic of the above implementation in Hyperledger, we obtain the practical latency and throughput shown in Table 3. All of three algorithms (i.e., Submit, Check, and Revoke) do not fail among respective 500

5. <http://vanet.eurecom.fr/>

6. <https://www.isi.edu/nsnam/ns/>

7. <https://github.com/colyn91/EBCPA>

4. <https://www.rinkeby.io>

Fig. 3. Simulation scenario with $1\text{km} \times 1\text{km}$.

test times. The average latency of Submit, Check, and Revoke is 0.25 second, 0.01 second, and 0.01 second respectively. This could be accepted in practice especially the Submit could be pre-executed periodically as mentioned in Section 4.4.1. Correspondingly, the throughput of them is 22.8 TPS, 314.5 TPS, and 335.8 TPS respectively, under setting sending rate as 25.1 TPS, 315.3 TPS, and 336.7 TPS respectively. Here, the “TPS” is abbreviated from “Transactions Per Second”. This means that the complete percentage of throughput in all these algorithms is more than 90%, which can satisfy the practical application requirements.

Computational and communication costs. The theoretical results are shown in Table 4, from which our EBCPA is more efficient than CertCoin [18] and BPAS [20] in all the phases in terms of both time cost and communication cost. This is due to that CertCoin requires a complex threshold ring signature and BPAS involves time-consuming ABE operations. In particular, the communication costs of CertCoin and BPAS are linear to r and l respectively (i.e., $168r +$

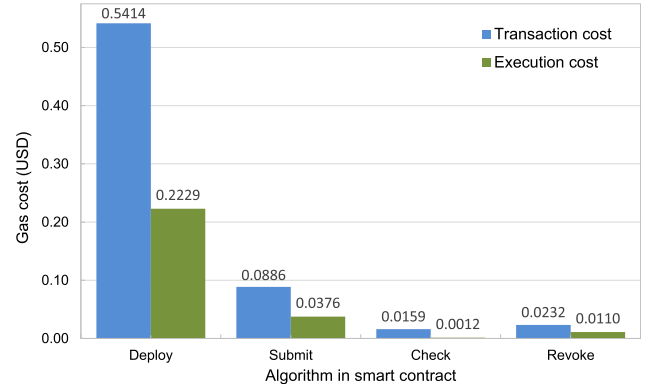


Fig. 4. Gas cost involved in our EBCPA protocol.

40 and $128l + 64$ bytes respectively, but that of EBCPA is a constant 392 bytes). While the message signing cost of EBCPA is similar to that of BCPPA [14] and communication overhead of the former is a little more, the message verification and tracing of EBCPA are more efficient than BCPPA. Especially, the tracing computation cost of BCPPA is linear to k (i.e., $k(T_{sm} + T_h)$), in contrast to that of our EBCPA is a constant (i.e., $T_{sm} + T_{pa}$). Therefore, the preliminary discussion shows that EBCPA is more practical than other comparing schemes.

The practical evaluation results (see Figs. 5 and 6) can further support the above conclusion. Our EBCPA indeed improves the computational efficiency especially the tracing and message verification. The time cost of tracing in EBCPA is with a constant 10.78 milliseconds, comparing to, that in CertCoin with 21.12 milliseconds, that in BCPPA with 106.01 milliseconds (if k is set as 10) and that in BPAS with a higher constant 87.56 milliseconds. Thus, under the setting

TABLE 3
Benchmark Results in the Hyperledger Test Network

Algorithm	Success	Fail	Send rate (TPS)	Max latency (second)	Min latency (second)	Average latency (second)	Throughput (TPS)
Submit	500	0	25.1	2.05	0.03	0.25	22.8
Check	500	0	315.3	0.03	0.00	0.01	314.5
Revoke	500	0	336.7	0.04	0.00	0.01	335.8

TABLE 4
Comparison With the Recent BCPPA [14] and BPAS [20] in VANETs

Item	CertCoin [18]	BCPPA [14]	BPAS [20]	EBCPA	
Time cost (millisecond)	Message Signing	$(2r - t)T_{sm} + u(r - t)T_{pa} + (2r + t)T_h + 2rT_{mm} + rT_{mi} + t(r - t)T_{enc} + T_{dec}$	$eth_calls + 3T_{sm} + T_{mi} + T_{mm} + T_h$	$T_{AEnc} + T_{sm} + T_{mm} + T_h$	$eth_calls + 3T_{sm} + T_{pa} + T_h + 2T_{mm}$
	One Message Verification	$3rT_{sm} + ruT_{pa} + (r + 1)T_h + (r - t)T_{enc}$	$tx_rtv + 4T_{sm} + 2T_{pa} + 2T_{mi} + 4T_{mm} + T_h$	$eth_calls + T_{ADec} + 2T_{sm} + T_{pa} + T_h$	$eth_calls + 3T_{sm} + 4T_{pa} + T_h$
	n -message Verification	$3nrT_{sm} + nruT_{pa} + n(r + 1)T_h + n(r - t)T_{enc}$	$n*tx_rtv + 4nT_{sm} + 2nT_{pa} + 2nT_{mi} + 4nT_{mm} + nT_h$	$n*eth_calls + nT_{ADec} + 2nT_{sm} + nT_{pa} + nT_h$	$n*eth_calls + 2nT_{sm} + 2T_{sm} + 4nT_{pa} + 3nT_{mm} + nT_h$
	Trace	$2T_{sm} + T_{pa} + T_{mi} + 2T_{mm} + T_h$	$k(T_{sm} + T_h)$	$T_{ADec} + T_{sm} + T_{mm} + T_h$	$T_{sm} + T_{pa}$
Communication cost (byte)	$r G + 3r Z_q + r ID + msg + ts = 168r + 40$	$3 Tid + 4 Z_q^* + msg + ts = 264$	$ c_{ABE} + G + Z_q^* + msg + ts = 128l + 200$	$4 G + 3 Z_q^* + msg + ts = 392$	

*We denote k as average searching number of one-time public key in [14], r, t, u as ring size, threshold size and bit length in [18] respectively. Here, we omit the time cost of retrieving transaction tx_rtv and querying smart contract eth_calls which are the scope of transmission latency.

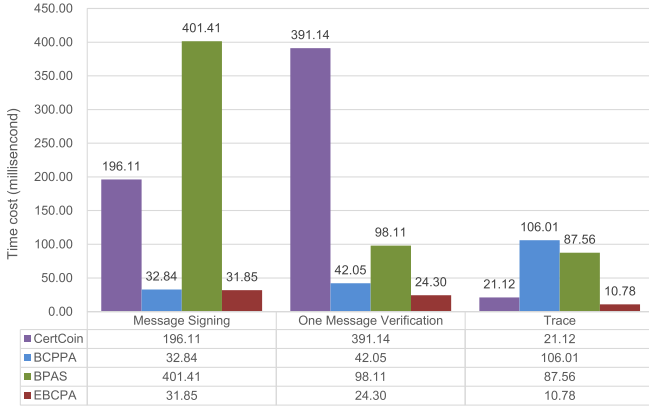
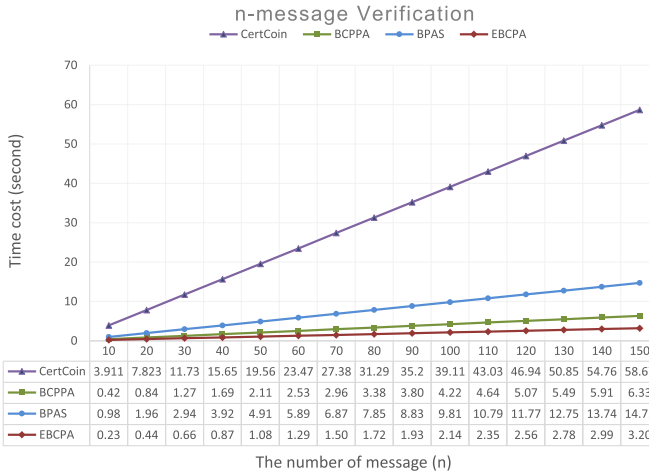


Fig. 5. Practical comparison results of different phases.

Fig. 6. Practical comparison of n -message verification efficiency.

of $k = 10$, our EBCPA can save about 48.95%, 89.83% and 87.68% computation costs of traceability than the CertCoin, BCPA and BPAS respectively. The message signing cost of CertCoin, BCPA, BPAS, and EBCPA is 196.11 milliseconds, 32.84 milliseconds, 401.41 milliseconds, and 31.85 milliseconds respectively (if r, t, u are set as 10, 5, 256 respectively). That is, our EBCPA can save at least 83.75%, 3.00% and 92.06% than CertCoin, BCPA and BPAS respectively in terms of message signing cost.

With respect to one message verification, our proposal involves 24.30 milliseconds, the CertCoin involves 391.14 milliseconds, the BCPA needs 42.05 milliseconds, and the BPAS costs 98.11 milliseconds. This shows that our EBCPA can save at least 93.78%, 42.21% and 75.23% (for one message verification time) comparing to the CertCoin, BCPA and BPAS respectively. In the n -message verification (see Fig. 6), all the comparing schemes involve linearly increasing time costs, but the growth rate of EBCPA is slower than that of others. The only compromise is that our EBCPA requires more communication overhead than BCPA (392 bytes versus 264 bytes). One can find that this small compromise could be accepted for a significant improvement of efficient signing, message verification, and tracing.

Message authentication delay and loss rate. Motivated by the average packet delay (APD) and packet loss ratio (PLR) defined in [44], we combine the aforementioned simulations to obtain the running results (see Figs. 7 and 8). In the former

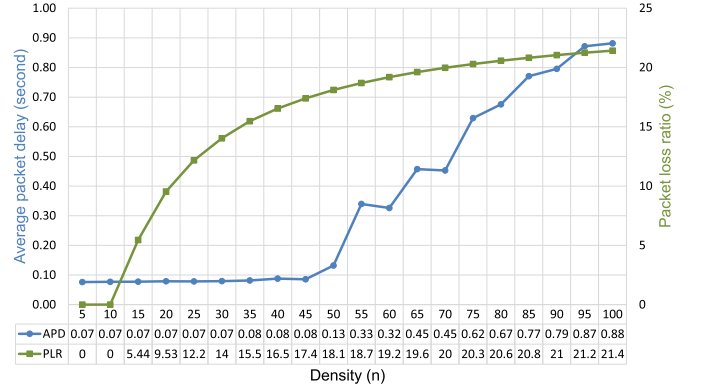


Fig. 7. The impact of vehicle density in APD and PLR.

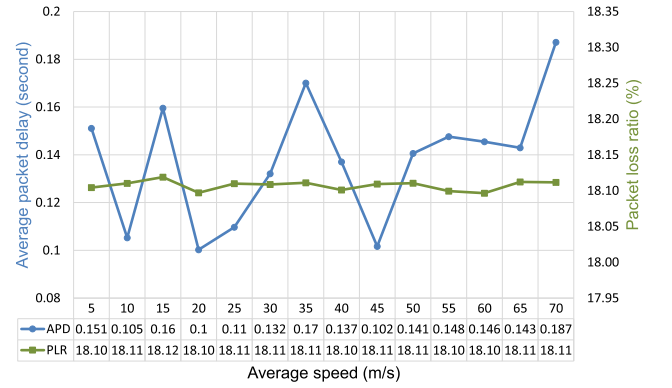


Fig. 8. The impact of vehicle average speed in APD and PLR.

simulation, the speed of vehicles is set as a fixed range (i.e., from 10 to 20 miles/second) and the density (i.e., the number of vehicles) changes from 5 to 100 (with setting the step as 5). As a result shown in Fig. 7, one can find that the APD only changes a little (less than 10 milliseconds) before the density increases to 50, but after that it increases quickly. Correspondingly, the PLR keeps increasing as the density increases, but it will be with a slow growth when the density is more than 50. This shows that the communication performance of VANETs (using our proposal) will be degraded when the density exceeds 50.

To evaluate the impact of vehicles' average speeds in APD and PLR, we fix the density as 50 and then change the average speed from 5 to 70. From the obtained results (see Fig. 8), as the average speed increases, the PLR is almost unchanged (about 18.1%) and the APD fluctuates largely (but the fluctuated span is less than 36 milliseconds). For one thing, the average speed of vehicles makes a little impact on the PLR within the same density, which fits the fact that only those data packets out of vehicles' range will be discarded. For another, different average speeds of vehicles may lead to that the distance among vehicles changes unpredictably and hence different APDs are caused.

7 CONCLUSION AND FUTURE WORK

This paper contributes to a more efficient blockchain-based privacy-preserving authentication (EBCPA) protocol for VANETs. We first design three system building blocks namely KeyDer scheme, SoK scheme and smart contract, following by the EBCPA. Moreover, we prove the security of

designed SoK scheme prior to analyzing the security of EBCPA. Finally, we implement it in the on-line Ethereum test network (Rinkeby), Hyperledger test network and VANETs simulation environment (via VanetMobiSim and NS-2), together with evaluating the communication and computational costs in our personal computer. The benchmarks and comparison demonstrate that our proposal is more suitable for blockchain-enabled VANETs from its comprehensive advantage especially the verification and traceability.

While our proposal has been demonstrated more efficient in the simulation environment, it remains indeterminacy in a real-word setting. In the future, we will further improve the efficiency of EBCPA via designing a specified blockchain (e.g., novel architecture, data structure, or consensus mechanism). In addition, we will evaluate our prototype based on a hybrid hardware and software implementation in the real-word VANETs.

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